

For problems 1 and 2, compute the following integral using the provided substitution. You should get the same answer (after all it will be the same integral) using different substitutions.

1) Compute the following integral by setting $u = x + 4$. You'll need a bit of algebra to figure out what to do with that x in the integrand.

$$\int x \sqrt{x+4} dx$$

$u = x+4$
 $x = u - 4$

$$\int (u-4) \sqrt{u} du$$

$$\int (u-4) u^{\frac{1}{2}} du$$

$$\int (u^{\frac{3}{2}} - 4u^{\frac{1}{2}}) du$$

power rule

$$\frac{2}{5} u^{\frac{5}{2}} - 4 \left(\frac{2}{3} u^{\frac{3}{2}} \right) + C$$

$$\frac{2}{5} u^{\frac{5}{2}} - \frac{8}{3} u^{\frac{3}{2}} + C$$

$$\frac{2}{5} (x+4)^{\frac{5}{2}} - \frac{8}{3} (x+4)^{\frac{3}{2}} + C$$

2) Compute the following integral by setting $u = \sqrt{x+4}$. (Hint: Find u^2 and then find du .)

$$u^2 = x+4 \text{ or } u = \sqrt{x+4}$$

$$\int x \sqrt{x+4} dx$$

$$2u du = dx$$

$$x = u^2 - 4$$

$$\int (u^2 - 4)(u)(2u du)$$

$$\int 2u^2(u^2 - 4) du$$

$$\int (2u^4 - 8u^2) du$$

power rule $f(x^n) = n x^{n-1}$

$$\frac{2}{5} u^5 - \frac{8}{3} u^3 + C$$

$$\frac{2}{5} (x+4)^{\frac{5}{2}} - \frac{8}{3} (x+4)^{\frac{3}{2}} + C$$

$$\frac{2}{5} (x+4)^{\frac{5}{2}} - \frac{8}{3} (x+4)^{\frac{3}{2}} + C$$

3) Compute the area of the region bounded by $y = 4$ and $y = x^2$ by partitioning the x -axis.

$$x^2 = 4$$

$$x = \pm 2$$

$$\text{Area} = 2 \int_{-2}^2 x \, dy$$

$$2 \int_0^4 \sqrt{y} \, dy$$

$$2 \int_0^4 y^{\frac{1}{2}} \, dy$$

$$2 \left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}} \right)_0^4$$

$$\frac{4}{3} (4^{\frac{3}{2}} - 0^{\frac{3}{2}})$$

$$\frac{4}{3} \sqrt{64}$$

$$\frac{4}{3} \cdot 8$$

$$\frac{32}{3} \text{ sq units}$$

4) Compute the area of the region bounded by $y = 4$, $y = 4x$, and $y = x^2$ by partitioning the y -axis.

$$y = 4x = x = \frac{y}{4}$$

$$y = x^2 = x = \pm \sqrt{y}$$

Desmos - first quadrant $x = \sqrt{y}$

$$\sqrt{y} = \frac{y}{4}$$

$$y = \frac{y^2}{16}$$

$$16y = y^2$$

$$y^2 - 16y = 0$$

$$y(y - 16) = 0$$

$$(y = 0, y = 16)$$

$$\int_0^4 \left(\sqrt{y} - \frac{y}{4} \right) dy$$

$$\int_0^4 \left(y^{\frac{1}{2}} - \frac{y}{4} \right)$$

power rule $f(x) = nx^{n-1}$

$$\left[\frac{2}{3} y^{\frac{3}{2}} - \frac{y^2}{8} \right]_0^4$$

$$\frac{2}{3} 4^{\frac{3}{2}} - \frac{4^2}{8}$$

$$\frac{2}{3} 8 - \frac{16}{8}$$

$$\frac{16}{3} - 2$$

$$\frac{16}{3} - \frac{6}{3}$$

$$\frac{10}{3}$$

$$\frac{2}{3} 4^{\frac{3}{2}} - \frac{0^2}{8}$$

$$\frac{2}{3} 0 - 0$$

$\emptyset$

$$\frac{10}{3} - 0 = \frac{10}{3}$$



5) Find the volume generated by rotating $y = \frac{1}{x}$ about the x -axis from $x = 1$ to $x = 5$

$$V = \pi \int_a^b [f(x)]^2 dx$$

$$V = \pi \int_1^5 \left[\frac{1}{x} \right]^2 dx$$

$$V = \pi \int_1^5 \frac{1}{x^2} dx$$

Power rule

$$\int x^{-2} dx = \frac{x^{-1}}{-1} = -\frac{1}{x}$$

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

$$V = \pi \left[-\frac{1}{x} \right]_1^5$$

$$-\frac{1}{5}, -\frac{1}{1}$$

$$V = \pi \left(-\frac{1}{5} - (-1) \right)$$

$$V = \pi \left(1 - \frac{1}{5} \right)$$

$$V = \pi \left(\frac{4}{5} \right)$$

$$\frac{4\pi}{5}$$

6) Find the volume generated by rotating $y = \frac{1}{x}$ about the y -axis from $y = 1$ to $y = 4$

$$V = \pi \int_c^d [g(y)]^2 dy$$

$$y = \frac{1}{x} \quad x = \frac{1}{y}$$

$$V = \pi \int_1^4 \left[\frac{1}{y} \right]^2 dy$$

$$V = \pi \int_1^4 [y^{-2}] dy$$

Power rule $f(x^n) = nx^{n-1}$

$$\int y^{-2} dy = \frac{y^{-1}}{-1} = -\frac{1}{y}$$

$$\int_a^b f(x) dx = Fx \Big|_a^b = F(b) - F(a)$$

$$V = \pi \left[-\frac{1}{y} \right]_1^4$$

$$-\frac{1}{4}, -\frac{1}{1}$$

$$V = \pi \left(-\frac{1}{4} - (-1) \right)$$

$$\pi \left(-\frac{1}{4} + 1 \right)$$

$$V = \frac{3\pi}{4}$$

7) Find the volume generated by rotating the region bounded by $y = 4 - x^2$ and $y = 1 - 0.25x^2$ about the x -axis.

2 equations, so washer method

$$V = \int_a^b \pi [(f(x))^2 - (g(x))^2] dx$$

$$\begin{aligned} 4 - x^2 &= 1 - 0.25x^2 \\ 3 &= .75x^2 \\ x^2 &= \frac{3}{.75} \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

$$V = \int_{-2}^2 \pi [(4 - x^2)^2 - (1 - 0.25x^2)^2] dx$$

$$\begin{aligned} &= \pi \int_{-2}^2 [(4 - x^2)(4 + x^2) - (1 - \frac{1}{4}x^2)(1 + \frac{1}{4}x^2)] dx \\ &= \pi \int_{-2}^2 [16 - 8x^2 + x^4 - (1 - \frac{1}{2}x^2 + \frac{1}{16}x^4)] dx \end{aligned}$$

$$V = \pi \int_{-2}^2 [15 - \frac{15}{2}x^2 + \frac{15}{16}x^4] dx$$

power rule

$$\left[15x - \frac{15}{6}x^3 + \frac{15}{80}x^5 \right]_{-2}^2$$

$$\left[15x - \frac{5}{2}x^3 + \frac{3}{16}x^5 \right]_{-2}^2$$

$$2\pi \left(15x - \frac{5}{2}x^3 + \frac{3}{16}x^5 \right)$$

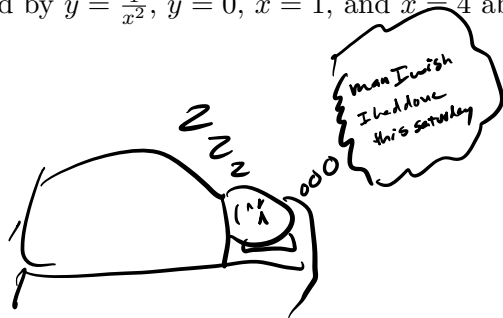
$$2\pi \left(15(2) - \frac{5}{2}(2)^3 + \frac{3}{16}(2)^5 \right) - 2\pi \left(15(-2) - \frac{5}{2}(-2)^3 + \frac{3}{16}(-2)^5 \right)$$

↓
calculator

$$2\pi (16)$$

$$32\pi$$

8) Find the volume generated by rotating the region bounded by $y = \frac{1}{x^2}$, $y = 0$, $x = 1$, and $x = 4$ about the line $y = 4$.



9) Let R be the region bound by the curves $x = 0$, $y = 0$ and $y = 4x^2$. Define a volume atop R such that each cross sectional slice drawn perpendicular to the y -axis is a semi circle whose diameter spans from the y -axis to the curve $y = 4x^2$. Find the volume of the solid atop R . I believe the answer 4π units³, if you get something different let's chat!

10) Let R be the region bound by the curves $y = x$ and $y = x^2$. Define a volume atop R such that each cross sectional slice drawn perpendicular to the x -axis is a rectangle whose base is the difference between the given curves and whose width is double that of the base. Find the volume of the solid atop R . I believe the answer is $\frac{1}{15}$ units³, if you get something different let's chat!